

STATE TECHNICAL UNIVERSITY

END SEMESTER EXAMINATION 2025-2026 | ANSWER BOOKLET

CANDIDATE ROLL NO: 2026-CS-PH-021
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EXAMINATION / COURSE: B.Tech Semester II
SUBJECT & CODE: Physics (CS 201)
BOOKLET SERIAL NO: BK-PH-80021

BARCODE SCAN TRACKING



BK-PH-80021

Q.No	Q1	Q2	Q3	Q4	Q5
Max	14	14	14	14	14
Marks					

START WRITING YOUR ANSWERS BELOW. DO NOT WRITE IN THE MARGINS.

Q.1

Ans I. Group and Phase Velocity:

- Phase velocity $v_p = \omega / k = E / p$.
- Group velocity $v_g = d\omega / dk = dE / dp$.
- For particle of mass m and velocity v :
 $p = mv$, $E = \frac{1}{2} m v^2 = p^2 / 2m$.
 $v_g = d(p^2/2m)/dp = 2p/2m = p/m = v$.
- So group velocity = particle velocity.

Q.2

Ans 2. Interference in Wave Optics:

- Superposition of light waves gives bright and dark fringes.
- Wavefront division: Young's Double slit experiment. Fringe width $\beta = \lambda D/d$.
- Amplitude division: Thin film interference and Newton's rings.
- Essential condition: Source must be coherent and monochromatic.

Q.3

Ans 3. Lasers and Fibre Optics:

- LASER = Light Amplification by Stimulated Emission of Radiation.
- Population Inversion: $N_2 > N_1$ using pumping.
- Optical fibers transmit light using Total Internal Reflection (TIR).
- Core has index n_1 , cladding has n_2 ($n_1 > n_2$).
- Numerical aperture $NA = \sqrt{n_1^2 - n_2^2}$.

Q.4

Ans 4. Electromagnetism Basics:

- Maxwell's 4 Equations:

1. $\text{div } \mathbf{E} = \rho / \epsilon$

2. $\text{div } \mathbf{B} = 0$

3. $\text{curl } \mathbf{E} = - d\mathbf{B}/dt$

4. $\text{curl } \mathbf{B} = \mu \mathbf{J} + \mu \epsilon d\mathbf{E}/dt$

- Displacement current $\mathbf{J}_d = \epsilon d\mathbf{E}/dt$.

- Tangential E-field is continuous at boundary ($E_{1t} = E_{2t}$).

- Poynting Vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ (Energy flow in W/m^2).

Q.5

Ans 5. Semiconductor Physics:

- Crystals: SC (52%), BCC (68%), FCC (74% packing).
- Band theory: Valence band and Conduction band separated by bandgap E_g .
- Intrinsic semiconductor: pure Si, $n = p = n_i$.
- n-type (donor doping, electrons majority), p-type (acceptor doping, holes majority).
- Direct bandgap used for LEDs and optical devices.

--- End of Answer Booklet (All 5 questions attempted) ---

Candidate Sign: Harsh